

Southeast University



Department of Computer Science & Engineering
Course No: CSE 1102

Course: Computer Fundamentals Lab

Submitted To

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INTRODUCTION

Telecommunication is the transmission of information over long distances using electronic signals and digital technology. From the telegraph in 1844 to today's 5G networks, it has transformed global communication.

Telecommunication is essential to modern life - enabling phone calls, internet access, television, and instant global communication. Without it, our digital world would not exist.

This project explores three main aspects:

1. Historical evolution from telegraph to 5G
2. Fundamental concepts of signal transmission
3. Modern technologies and their real-world applications

MAIN DISCUSSION

1. HISTORY AND EVOLUTION

Key milestones in telecommunication:

- 1844 - Telegraph (Samuel Morse): First long-distance communication using electrical signals
- 1876 - Telephone (Alexander Graham Bell): Enabled voice communication
- 1906 - Radio: Wireless communication begins
- 1927 - Television: Visual transmission
- 1973 - Mobile Phones: Beginning of mobile era
- 1991 - 2G Networks: Digital cellular networks
- 2009 - 4G/LTE: High-speed mobile internet
- 2020 - 5G Networks: Ultra-fast, low-latency networks

2. HOW TELECOMMUNICATION WORKS

Three basic components:

- A) Transmitter: Converts information into electronic signals
- B) Transmission Medium: The channel (wired or wireless) carrying signals
- C) Receiver: Converts signals back into usable information

Basic Elements:

- Sender: Initiates communication
- Signal: Electronic representation of information

- Channel: Physical path for signal transmission
- Receiver: Decodes the signal

3. TYPES OF SYSTEMS

- Point-to-Point: Direct communication between two devices (e.g., phone calls)
- Broadcast: One transmitter to many receivers (e.g., radio, TV)
- Multiplex: Multiple simultaneous users (e.g., cellular networks)

4. TECHNOLOGY COMPARISON

<i>Technology</i>	<i>Year</i>	<i>Download Speed</i>	<i>Upload Speed</i>	<i>Latency</i>	<i>Users(M)</i>
2G(GSM)	1991	0.064 Mbps	0.064 Mbps	500 ms	3500
3G(UMTC)	2002	2 Mbps	1 Mbps	100 ms	2000
4G (LTE)	2009	100 Mbps	50 Mbps	50 ms	5000
5G(NR) 2020		10000 Mbps	5000 Mbps	ms	600
WIFI-6 2019		1200 Mbps	800 Mbps	20 ms	1500

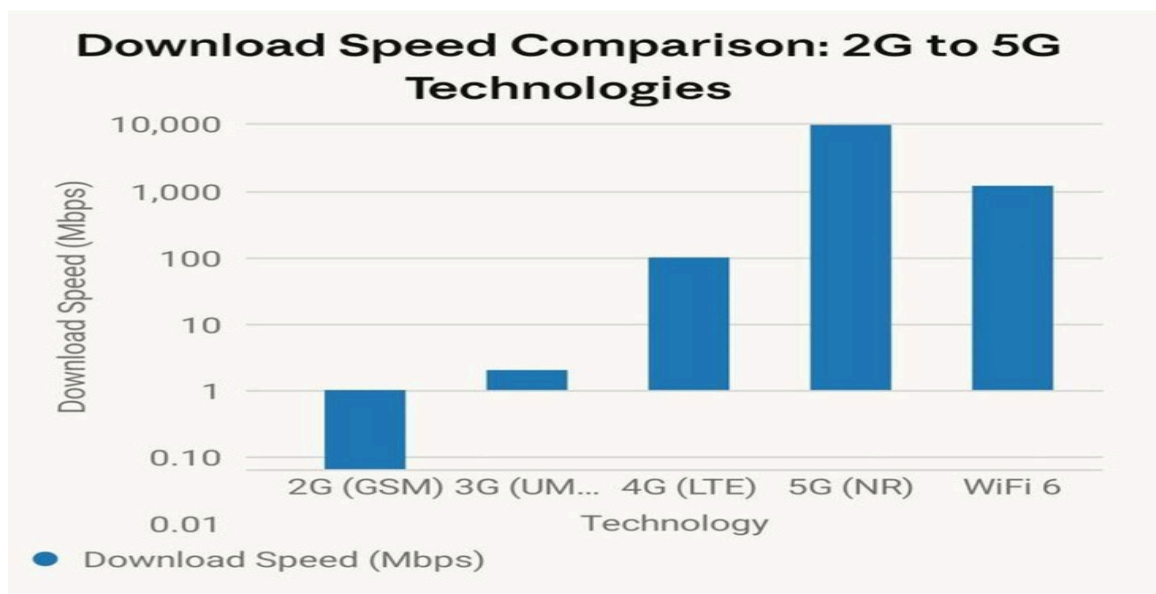


Chart Explanation:

The chart shows exponential growth in download speeds from 2G to 5G. Key observations:

- 5G is 156,000 times faster than 2G

- Speed doubles roughly every 10-15 years
- Latency (delay) decreased from 500ms to 1ms
- 4G has highest user adoption globally • 5G still in early adoption due to cost

CONCLUSION

Telecommunication has evolved from telegraph to 5G networks, fundamentally transforming human communication. From enabling instant global connectivity to supporting emerging technologies like IoT and AI, telecommunication infrastructure forms the backbone of modern civilization.

The exponential improvements in speed, latency, and reliability demonstrate continuous innovation. As we move toward 6G and beyond, understanding these principles becomes increasingly important for our digital future.

REFERENCES

1. International Telecommunication Union (ITU). "5G Technology Guide"
2. Cisco Systems. "Mobile Network Evolution Report"
3. GSMArena. "Technology Specifications Database"
4. Wikipedia. "Telecommunication"
5. IEEE. "Communication Systems Standards"

Team Member Contribution

NAME	ID NUMBER	CONTRIBUTION
Md Sajid	2026000000239	Created Excel spreadsheet with data, generated chart ,analyzed trends. Spent -12 Hours
Abir Ullah Chowdhury	2026000000214	Designed presentation slides, set up GitHub repository, ensuring proper formatting, and content organization . Spent-12 Hours
Shahabuddin Chowdhury	2026000000315	Wrote introduction,main discussion,Conclusion, Reasearched history and fundamentals.Spent-9 Hours
Md. Fardin Khan	2026000000220	Manage all presentation slide design details and final refinements , visual consistency, content and wrote README documents. Spent- 12 Hours

AI USEAGE DOCUMENTATION

<i>Tool</i>	<i>Purpose</i>	<i>Used</i>	<i>Modification</i>
ChatGPT	Concept learning	Yes	Rewrote in own word
Google Gemini	Data interpretation	Yes	Added own analysis
ChatGPT	Grammar check	Partial	Only spelling fixes
Google AI	Spell check	Minimal	Auto correct only